<u>Industry Advisory Board Meeting Notes</u>

July 22nd, 2019

Agenda*

Sign In **9:00am**

Meeting called to order **9:30am**

Opening remarks – Dr. Zdunek **9:30am**

Agenda reviewed

State of the Department – Dr. Berry **9:45am**

- Student success
- EIB (New Building)
  - Structure
  - Offices/Labs
  - Outlines
  - Video Presentation (EIB)
  - Connecting with Campus Resources
  - Unit Operations Lab
- Concentrations
- Highlights of Chemical Engineering changes and academic plans
- Graduate and Undergraduate
  - Moved up in rakings (US News)
  - Undergraduates
    - Higher locally, lower nationally (field)
  - Graduate Funds
    - We have secured higher outside funds (non-state funds)
  - Research areas
    - Cancer Research
    - Polymers
    - Nano Assembly
    - Artificial Synthesis
    - Drug Creation
    - DoD Funding
  - Student Highlights
    - Graduate Recognitions
    - Company affiliations
    - Chem E Car
- New Position
  - Full Professor
  - Fellowship award conjunction
- New acquisition
  - XPS

- Publications
    - Media
        - Printed
        - Social
        - Conventional
- Wrap up
    - Q&A
        - Call to questions
            - Vote
                - Will board accept MS students for one semester to train/teach?
                    - Further exploration needed

ABET Update and Program Objectives – Dr. Alan Zdunek **10:30am**

- Timeline to Fall 2020 Assessment
- Criterion to PASS ABET Assessment
- Self-Study Report
- Overview and Ongoing Tasks
- Assessment
    - New Educational Objectives
    - Student Learning Outcomes
    - Program Curriculum
        - Changes (Proposal)
    - Course Portfolios
    - Course Self-Improvement
    - Senior Exit Survey and Exam
    - Ethics Exam
        - Execution
        - New approach?
            - Having it a part of the curriculum?
            - Develop a new course? (through dept./college/university)
            - Incorporate it into existing courses?
            - Professional Development?
            - Discussions
- Review and IAB Approval Current Program Educational Objective
    - Vote
        - Proposed changes to Objectives
        - Depth – Creative design, conceptual innovation, etc.
    - Approved Changes

Undergraduate Curriculum Changes **11:15am**

- Changes from electives to permanent courses
- Other course selections
    - Teach corrosion?
    - Computational language discussion

        - MATLAB
        - Python
- CS 109 Discussion
    - Should we work in Statistics course?
    - C+/++ needed?
    - CHE courses do cover regression, deviation, etc.
- Chemistry
    - Organic Chemistry
        - Connect Orgo properties with Chemical Engineering

****Discussion****

Q&A

- Process design (concentrations changes)
- Student Placement
    - Getting data from alumni and graduates
- Open Position
    - Faculty, Full-Professorship
- NSF Lab Materials
    - Mostly DoD
    - DOE
        - Kim
        - Chaplin
- Internship Programs
    - GPIP
    - Explore other options
    - Summer Courses
        - UIC Business Model?
    - Partnerships
    - External PhD program
    - Certificates vs. Degrees
    - CPT/OPT
        - Preplace?
        - Immigration status

****Transition****

New Areas of Concentrations **10:15am**

- Basic Knowledge vs. Going towards a more specific area
    - Discussion
    - Reverse teaching; integrated process from end product to basic elements
    - Conclusion

            - Course should be Molecular and Engineering Materials not just Molecular or Polymer
- Concentrations – Discussions
    - Changes proposed to Polymer Concentrations (Change Econ to System and Industrial)
    - Changes in required courses
    - **Board members will review and submit their final suggestions**
        - **Deadline – August 23rd**

Data Collecting

- Change collection (break down of respondents; profession, level of education, etc.)
- Change rankings of importance (3)

Department Needs

- Endowed Professorship
- Endowed Chair
- Computer upgrades
- Unit Ops lab
- Experiment needs
- Chem E Car Fellowships
- Faculty Start Up
    - Molecular
    - Polymer
    - Bio Med
    - Complex Fluids
    - Nanotechnology

Next Meeting Items

Next Meeting

- Spring 2020
- Have students present
- Action Items
- Meeting notes
- Follow ups

Meeting Adjourned **12:30pm**

Appendix

Members

Alan Zdunek	X
Betul Bilgin	X
Christos Takoudis	
Dennis O'Brian	X
Diane Graziano	X
Gang Cheng	X
Gavin Towler	X
Lewis Wedgewood	X
Meenesh Singh	X
Michael Caracotsios	X
Said Al-Hallaj	X
Sangil Kim	
Sanjay Behura	X
Soundbar Kumaran	
Tarun Chandra	X
Vikas Berry	X
Ying Liu	

AGENDA

Date: Monday, July 22, 2019
Time: 9:00am-1:30pm
Location: Science and Engineering Office (SEO) Building, 851 South Morgan St., Room 1000 (10th Floor)

9:00-9:30 Arrival and Check-in

9:30-9:45 Welcome by Professor Zdunek

9:45-10:30 State of the Department presentation by Professor Berry

10:30-11:15 ABET update and program objectives discussion

11:15 - 12:15 Discussion of undergraduate curriculum changes

12:15 – 12:30 Wrap-up

12:30-1:30 Luncheon

1:30-2:00 Break before Engineering Innovation Building Celebration

2:00 – 4:00 Engineering Innovation Building Ribbon Cutting Celebration (Optional)

CHANGES TO THE DEGREE-REQUIREMENTS AND INTRODUCTION OF NEW COURSES AND CONCENTRATION AREAS (INCLUDING MOLECULAR ENGINEERING)

The Department of Chemical Engineering

University of Illinois at Chicago

The department of chemical engineering will be submitting a proposal to modify its BS-Degree requirements with the introduction of new courses and several concentrations, including molecular engineering.

THE CHANGES TO THE CURRICULUM

The following are the proposed changes to the curriculum in chemical engineering:

1. Introduction of four new courses:
    a. CHE 2XX: Molecular Systems in Chemical Engineering (Core Course)
    b. CHE 3XX: Polymer Science (Core Course) with CHE301 and CHE311 as prerequisite
    c. CHE 4X1: Molecular and Macromolecular Engineering (Elective)
    d. CHE 4X2: Nanoscale Systems in Chemical Engineering (Elective)
    e. CHE 4X3: Colloidal and Interfacial Phenomena (Elective)
2. Choice between CHE 3XX and CHEM 346 (Physical Chemistry - II) as a required course for Bachelor of Science in Chemical Engineering degree.
3. Choice between CHE 2XX and CME 260 towards BS degree
4. Convert courses into 500 Level Courses
    a. CHE 410 (Transport Phenomena) into CHE5XX (Advanced Transport Phenomena)
    b. CHE 431 into CHE5XX
    c. CHE 445 into CHE5XX
5. Addition of new concentrations
    a. Concentration in Process Simulations
    b. Concentration in Polymer and Molecular Engineering
    c. Concentration in Nanotechnology
    d. Concentration in Energy and Environment
    e. Concentration in Intellectual Property and Entrepreneurship

CONCENTRATION IN PROCESS SIMULATIONS

1. REQUIRED COURSES
    a. CHE 2XX (between CHE 2XX and CME260)
    b. CHE 3XX (Core Class to be taken by all anyways)
    c. CHE 433. Process Simulation with Aspen Plus (as Technical Elective)
2. ELECTIVE COURSES
    a. Elective from outside ChE-rubric from the following list:
        i. ECON 120 and ECON 220 Theory and Applications of Microeconomics (+3 credits)
        ii. IE 201 Financial Engineering
        iii. MATH 310 Applied Linear Algebra
        iv. MATH 419 Models in Applied Mathematics

<u>TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 3 credit hours, if ECON 120/220 is taken; otherwise 0 additional credits</u>

CONCENTRATION IN POLYMER AND MOLECULAR ENGINEERING

What is Molecular Engineering

Molecular engineering is an interdisciplinary field of study focused on design, characterization, synthesis, assembly, analysis, simulation and applications of molecular-scale systems. Molecular engineering also includes the study of interaction of different molecular systems and molecular assembly to achieve desired properties for specific functions. This process broadly falls under the category of “bottom-up” design.

Molecular engineering includes polymer engineering, interface engineering, pharmaceutical engineering, energy harvesting, environmental engineering, nanotechnology, synthetic biology, molecular machines, drug-delivery, self-assembly, and molecular electronics. These areas are already being pursued by the department of chemical engineering.

Design Elements: While engineering principles are generally applied to combine different materials to design engineering systems; molecular engineering designs molecular systems from the atomic scale by applying chemical and physical principles to control their system-level properties.

What is the relationship between molecular engineering and chemical engineering?

Molecular engineering is closely related to chemical engineering. Chemical engineers apply chemistry and physics at macro-scale to design processes for large-scale manufacturing of chemicals; while molecular engineers apply chemistry and physics at atomic/molecular scale to design molecules, molecular systems, and molecular assemblies with desired attributes. Since the department of chemical engineering already includes detailed curriculum on chemical processes, the addition of molecular engineering will complete the story from the origin of the molecule to its production at large scale with desired properties for their intended applications.

1. REQUIRED COURSES
    a. CHE 2XX (between CHE 2XX and CME260)
    b. CHE 3XX Polymer Science
    c. CHE 4X1 Molecular and Macromolecular Engineering (as Technical Elective).
2. ELECTIVE COURSES
    a. Elective from outside ChE-rubric from the following list:
        i. PHYS 450/BIOE 450. Molecular Biophysics of the Cell
        ii. PHYS 461 Thermal and Statistical Physics
        iii. BIOE 405. Atomic and Molecular Nanotechnology
        iv. CHEM 458. Biotechnology and Drug Discovery
        v. CHEM 448 Statistical Thermodynamics
        vi. CHEM 452/454 Biochemistry
        vii. PHAR 422. Fundamentals of Drug Action.
        viii. PHAR 423. Biomedicinal Chemistry
    b. Second Elective from the following list:
        i. CHE 4X2 Nanoscale Systems in Chemical Engineering
        ii. CHE 4X3: Colloidal and Interfacial Phenomena
        iii. CHE 494: 2D Nanomaterials

iv. CHE 494: Electrochemistry
v. CHE 425. Nanotechnology for Pharmaceutical Applications
vi. CHE 494: Macromolecular and Supramolecular Synthesis
vii. CHE 438: Computational Molecular Modeling
viii. CHE 494: Soft-Matter and Complex Fluids
ix. CHE 440: Non-Newtonian Fluids
x. CHE 494: Nano and Micro Engineered Membranes
xi. CHE 392: Undergraduate Research on Topics of Polymer and Molecular or Engineering

TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 3

CONCENTRATION IN NANOTECHNOLOGY

1. REQUIRED COURSES
    a. CHE 2XX Molecular Systems in Chemical Engineering (between CHE 2XX and CME260)
    b. CHE 3XX: Polymer Science (Core Course)
    c. CHE 4X2 Nanoscale Systems in Chemical Engineering (as Technical Elective)
2. ELECTIVE COURSES
    a. Elective from outside ChE-rubric from the following list:
        i. ECE 346. Solid State Device Theory.
        ii. ECE 440. Nanoelectronics.
        iii. ECE 449. Microdevices and Micromachining Technology.
        iv. BIOE 405. Atomic and Molecular Nanotechnology
        v. BIOE 485. Nanobiosensors
        vi. CHEM 458. Biotechnology and Drug Discovery
        vii. ME 418. Transport Phenomena in Nanotechnology.
        viii. PHYS 431: Modern Physics: Condensed Matter.
    b. Second Elective from the following list:
        i. CHE 494: 2D Nanomaterials
        ii. CHE 494: Nanomaterials for Energy Applications
        iii. CHE 494: Colloidal and Interfacial Phenomena
        iv. CHE 425. Nanotechnology for Pharmaceutical Applications.
        v. CHE 494 (With Topics related to Nanotechnology)
        vi. CHE 392: Undergraduate Research on Topics Thematically linked to Nanotechnology

<u>TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 3</u>

CONCENTRATION IN ENERGY AND ENVIRONMENT

1. REQUIRED COURSES
    a. CHE 2XX (between CHE 2XX and CME260)
    b. CHE 3XX (Core Class to be taken by all anyways)
    c. CHE 451. Renewable Energy Technologies (Technical Elective)
    d. CME 322. Environmental Engineering. (Additional)
    e. CME 421. Water Treatment Design (Elective from outside ChE-rubric).
2. ELECTIVE COURSES
    a. Second Elective from the following list:
        i. CHE 494: Electrochemistry
        ii. CHE 494: Nano and Micro Engineered Membranes
        iii. CHE 392: Undergraduate Research on Topics Thematically linked to Sustainability, Electrochemistry, Water Science, and Renewable Resources.
        iv. CHE 450. Air Pollution Engineering

<u>TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 6</u>

CONCENTRATION IN ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY

1. REQUIRED COURSES
    a. CHE 2XX (between CHE 2XX and CME260)
    b. CHE 3XX (Core Class to be taken by all anyways)
    c. CHE 494. Entrepreneurship in Chemical Engineering (as Technical Elective)
2. ELECTIVE COURSES
    a. Elective from outside ChE-rubric from the following list:
        i. ENGR 401. Engineering Management
        ii. ENGR 402. Intellectual Property Law
        iii. ENGR 404. Entrepreneurship.

<u>TOTAL ADDITIONAL CREDITS FOR THE CONCENTRATION = 0</u>

SYLLABUS OF THE NEW COURSES

1. CHE 2XX Molecular Systems in Chemical Engineering

Prerequisites: Organic chemistry CHEM 232

The objective of this course is to provide a general background on molecular systems for undergraduate level students. Fundamental topics that will be introduced include chemical bonding; synthesis mechanisms; nanoparticle nucleation; molecular characterization; the mechanical, electrical, and optical properties of molecular systems; crystal structure and defects; diffusion and Brownian motion; interfacial phenomena; phase diagrams of molecular systems; and applications of synthetic macromolecules, biomacromolecules and nanomaterials. This course covers examples of molecular systems used in energy, medicine, environmental and other technology applications.

ABET learning outcomes

1. Ability to apply knowledge of mathematics, science, and engineering to solve material problems related to chemical engineering.
2. Development of an understanding of professional and ethical responsibility.
3. Broad education necessary to understand the impact of engineering and scientific solutions in a global, economic, environmental, and societal context.

Course learning outcomes

One of the main objectives of the course is to familiarize the students with the fundamental concepts in molecular engineering, materials chemistry, and molecular systems, which will be used as background knowledge for the understanding of specialized problems in the field of Chemical Engineering. At the end, students will be able to

1. Understand the fundamental and essential concepts used in materials chemistry and molecular systems

2. Classify the molecular systems, nanomaterials, and materials systems.

3. Know the basic methods to synthesize and characterize chemicals and materials systems

4. Understand the basic properties that characterize the behavior of molecular systems

5. Select appropriate type of material for specific application

6. Offer different approaches to modify structure/microstructure in order to get desired properties

Course learning outcome assessment methods

1. Weekly homework assignment will assess student's mastery of the basic principles and concepts of molecular systems.
2. A mid-term exam, a final exam and six quizzes will assess students' ability to design, characterize, and evaluate molecular systems, nanomaterials, and materials systems.

3. Two individual paper will assess the students' ability of identifying important problems and discussing the challenges and possible solutions.
4. One team presentation will assess the students' ability of teamwork.

Purpose of this course in relation to overall curriculum and relationships to other courses offered by primary unit:

This course is designed to provide more comprehensive knowledge of molecular systems and material science in many emerging fields of Chemical Engineering. For regular Chemical Engineering undergraduates, this course can serve as an alternative course of CME 260 Properties of Materials that is offered by the Department of Civil and Materials Engineering. This course covers both materials chemistry and physics of bulk materials and nanomaterials for a broad range of applications. This course is a core course for Chemical Engineering undergraduates and for those who opt for concentration in Molecular Engineering. This course does not overlap with any other ChE courses, but it is a prerequisite course for the other advanced courses for Molecular Engineering Concentration.

Relationship of this course to similar courses offered by other academic units:

There is minor overlap with CME 260 Properties of Materials.

1. Introduction (3 hours)
    a. The Science and Engineering of Large molecules, Small Particles, Self-Assembled Structures: Examples from Biology, and Industry
    b. Chemical vs Physical Bond. Covalent Bond. Hydrogen Bond. Hydrophobic Interactions.
    c. Intermolecular Forces. Cohesive Forces. Interfacial Tension.
    d. Properties of Interfaces and Double-Layers
2. Materials Properties (3 hours)
    a. The mechanical properties of molecular systems
    b. The electrical, magnetic, and optical properties of molecular systems
    c. Crystal structure and defects, diffusion, and phase diagrams of molecular systems
3. Polymer Synthesis and Characterization (12 hours)
    a. Basic concepts in polymer science, Chemical bonding in polymers – ionic (ionomers), covalent, coordinate, metallic (Metallocene polymers), hydrogen bonding, Stereochemistry of polymers
    b. Monomer structure and polymerizability, Methods of polymerization, Chain reaction (Addition) polymerization and Kinetics of free radical addition polymerization, Ionic and coordination chain (addition) polymerization

c. Controlled polymerization methods: Atom Transfer Radical Polymerization (ATRP), Group Transfer Polymerization (GTP), Reversible-addition-fragmentation chain-transfer polymerization (RAFT), Nitroxide mediated polymerization (NMD).

d. Copolymerization and Step reaction (condensation) polymerization

e. Morphology and order in crystalline polymers, Polymer structure and physical properties: The crystalline melting point, the glass transition, Factors affecting Tm and Tg. Determination of Tg by a. Dilatometer, b. TMA and c. DSC, Properties involving large deformations, properties involving small deformations, property requirements and polymer utilization.

f. Polymer chains characterization and Analytical Chemistry of polymers

g. Rheology and mechanical properties of polymers, Mechanical models – Maxwell and voigt Boltzmann's superposition principles. Kinetic theory of rubber elasticity. The glassy state and the glass transaction, dynamic mechanical testing, relaxation spectrum, frequency dependent visco-elastic behavior stress – strain behavior of elastomers, the mechanical properties of crystalline polymers.

h. Polymer Processing (Injection Molding, Extrusion, Fiber Manufacture, Membrane Design)

4. Synthesis and Characterization of Nanomaterials (9 hours)

   a. Bottom-Up Approaches (ALD, CVD, MBE)
   b. Top-Down Approaches (Lithography)
   c. Vapor Condensation
   d. 2D nanomaterials growth
   e. Nanoparticle synthesis
   f. MOF Synthesis
   g. Characterization of nanomaterials (microscopy, spectroscopy)

5. Engineering Biomacromolecules (9 hours)

   a. Introduction of biomolecular Engineering and Basic biochemistry and molecular biology

   b. Polymerase Chain Reaction (PCR) basics, site-directed and random mutagenesis, display techniques, DNA sequencing

   c. DNA and Protein gel electrophoresis, cutting and splicing DNA, protein expression, DNA and RNA hybridization techniques, and Western blots

   d. Different protein design approaches, including knowledge-based design, computational protein design, and directed evolution, that are commonly used for protein engineering. Examples of engineered proteins with novel structural and functional properties are extensively discussed to illustrate how design principles are applied to real life problems.

   e. Expression systems of biomolecules

f. Basic principles of fermentation processes, scaling-up

g. Product recovery and separation (cell disruption, sedimentation, filtration/centrifugation, solvent extraction, adsorption, precipitation), product purification (adsorption, ultrafiltration, crystallization) and final separation and advances in bioseparation strategies.

h. Biotechnology or biomedical applications of biomolecules

6. Self-Assembled Materials: Driving Forces and Characterization (3 hours)
    a. Introduction to Self-Assembly of Amphiphilic Molecules
    b. Driven Assembly
    c. Mixing, Demixing and Spinodal Decomposition
    d. Ordering and Hierarchical Assembly
    e. Self-Assembly in Biology
7. Engineering Functional Materials (6 hours)
    a. Nanocomposites
    b. Special Topics (Liquid Crystalline Materials and Designing Displays; Micellar Synthesis of Nanomaterials; Design of Membranes for Energy and Water Purification; Design of Polymer-based Photovoltaics; Electronic Applications of 2D Materials; nanomedicine)

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2. CHE 3XX Polymer Science in Chemical Engineering

Prerequisites: CHE 2XX (Molecular Systems in Chemical Engineering) and Organic chemistry CHEM 234

Objectives: This course provides a broad overview of polymer science and engineering. The main objective is to introduce polymers as an engineering material and emphasize the basic concepts of their nature, production and properties. Polymers are introduced at three levels; namely, the molecular level, the micro level, and macro-level. The course also aims at introducing the students to the synthesis and structure of polymeric materials, the crystalline and glassy states, solution and melt properties, phase behavior, mechanical and rheological properties. Through knowledge of all three levels, student can understand and predict the properties of various polymers and their performance in different products.

Textbooks

(1) Polymer Chemistry, Second Edition / Edition 2 by Paul C. Hiemenz, Timothy P. Lodge, Tim Lodge

(2) Polymer Physics (Chemistry), 1st Edition by M. Rubinstein, Ralph H. Colby

ABET learning outcomes

1. Ability to apply knowledge of mathematics, science, and engineering to solve material problems related to chemical engineering.

Course learning outcomes

One of the main objectives of the course is to familiarize the students with the fundamental concepts

of Materials Science and Engineering which will be used as background knowledge for the

understanding of specialized problems in the field of Chemical Engineering. At the end, students will be able to

1. Understand the fundamental and essential concepts used in polymeric materials for Chemical Engineering applications.

2. Know the detailed methods to synthesize and characterize the polymeric materials

3. Understand the properties that characterize the behavior of polymeric materials

4. Select appropriate type of polymeric material for specific application

5. Offer different approaches to modify structure/microstructure of polymeric materials in order to get desired properties

Course learning outcome assessment methods

1. biweekly homework assignment will assess student's mastery of the basic principles and concepts of molecular systems.

2. A mid-term exam and a final exam will assess students' ability to design, characterize, and evaluate materials.

3. Four individual papers and One presentation will assess the students' ability of identifying important problems and discussing the challenges and possible solutions.

Purpose of this course in relation to overall curriculum and relationships to other courses offered by primary unit:

This course does not overlap with any other ChE courses. It is a core course for Molecular Engineering Concentration. This course is designed to provide more comprehensive knowledge of molecular systems and material science in many emerging fields of Chemical Engineering.

Relationship of this course to similar courses offered by other academic units:

No overlap

Contents

1. **Introduction of polymer science (3 hours)**

    a. Concept of macromolecules.

    b. Degree of polymerization, Concept of molecular mass, polydispersity, number average and weight average, viscosity average molecular weight and their statistical equations, molecular weight distribution in linear polymers (step growth and chain polymers), Nomenclature of polymers.

    c. Basic concepts in polymer science.

    d. Different ways in classification of polymers depending on –

        a) The origin (natural, Semisynthetic, synthetic etc.)

        b) The structure (linear, branched, network, hyperbranched, dendrimer.)

        c) The type of atom in the main chain (homochain, heterochain).

        d) The formation (condensation, addition).

        e) Homopolymers, copolymers.

        f) The behavior on application of heat and pressure (thermoplastic and Thermosetting).

        g) The form and application (plastics, fiber. elastomers and resin).

    e. Chemical bonding in polymers – ionic (ionomers), covalent, coordinate, metallic (Metallocene polymers), hydrogen bonding.

    f. Stereochemistry of polymers, Introduction to two types of polymerization

    g. Reactions viz. condensation and addition polymerization (without detailed

    h. mechanism and derivations)

    i. Monomer structure and polymerizability.

    j. Concept of functionality. Writing the structure of the polymer formed for a given monomer and its classification. Raw materials for monomers with specific example: acrylonitrile, vinyl, chloride, methyl methacrylate, isobutylene, isoprene, styrene, hexamethylene diamine and adipic acid, caprolactam, ethylene oxide and sebasic acid, ethylene glycol and terephthalic acid and their Polymerization reactions.

2. **Methods of polymerization (12 hours)**

a. Methods of polymerization: Bulk polymerization, Solution polymerization, Emulsion polymerization, Suspension polymerization, ROP, Interfacial polymerization, Melt polycondensation, Solution polycondensation.

b. Chain reaction (Addition) polymerization

c. Free radical addition polymerization mechanism of vinyl polymerization, generation of free radicals, initiation, propagation, termination, chain transfer inhibition of retardation, configuration of monomer units in vinyl polymer chains.

d. Kinetics of free radical addition polymerization – experimental determination of rate constants, derivations for rate expressions and expressions for kinetic chain length and hence degree of polymerization. Control of molecular weight by transfer, molecular weight and its distribution. Thermodynamics of free radical polymerization, effect of temp and pressure, enthalpies, entropies, free

e. energies, activation energies o polymerization.

f. Ionic and coordination chain (addition) polymerization common features of two types of ionic polymerization, Mechanism of cationic polymerization, expressions for overall rate of polymerization and the number average degree of polymerization. Mechanism of anionic, polymerization, expressions for overall rate of polymerization and the average degree of polymerization, living polymers. Mechanism of coordination polymerization – Ziegler-Natta catalysts, expressions for overall rate of polymerization. Ring opening polymerization-mechanism of polymerization of cyclic ethers, cyclic amides and cyclosiloxanes.

g. Copolymerization – types of copolymerization- the copolymer composition equation, monomer reactivity rations, rate of copolymerization, composition of copolymers, variation of copolymer composition with conversion, mechanisms of copolymerization, block and graft copolymers.

h. Controlled polymerization methods, viz, Nitroxide mediated polymerization (NMD),Atom Transfer Radical Polymerization (ATRP), Group Transfer Polymerization (GTP), Reversible-addition-fragmentation chain-transfer polymerization (RAFT).

i. Step reaction (condensation) polymerization – Mechanism of step reaction polymerization, carbonyl addition elimination, carbonyl addition – substitution, nucleophilic substitution, and aromatic electrophilic substitution. Kinetics of step reaction polymerization, reactivity and molecular size. Kinetic expressions for polymerization in absence and in presence of a catalyst. Statistics of linear step reaction polymerization – number distribution and weight distribution functions, molecular weight control, Polyfunctional step reaction polymerization, prediction of gel point, its experimental observation, molecular wt. Distribution in – 3 D step reaction.

3. Analytical Chemistry of polymers (3 hours)

a. Test procedures and brief discussion on following terms:

b. Tensile strength, Impact strength, Elongation at break, water resistance, hardness, Heat distortion temperature, brittleness, Flexural strength, and Tm, Tg.

c. Measurement of molecular weight and size, Molecular weight determination of Polymer: end group analysis, colligative property measurement; light scattering,

d. Gel permeation chromatography, Fractionation of polymers by solubility, viscometric method, ultracentrifugation, and poly –electrolytes.

4. Physical Chemistry of polymers (12 hours)

a. Morphology and order in crystalline polymers: Configurations of polymer chains, crystal structures of polymers, Morphology of polymer single crystals, structure of polymers crystallized from melt and solution, crystallization processes and kinetics, orientation and drawing.

b. Polymer structure and physical properties: The crystalline melting point, the glass transition, Factors affecting Tm and Tg. Determination of Tg by a. Dilatometer, b. TMA and c. DSC, Properties involving large deformations, properties involving small deformations, property requirements and polymer utilization.

c. Polymer chains and their characterization.

d. Polymer solutions – Criteria of polymer solubility, conformations of dissolved polymer chain, stages and thermodynamics of polymer solutions nature (size and shape) of polymer in solutions, theta temperature, viscosity of dilute solution, phase separation in polymer solutions, moderately highly concentrated solutions.

e. Radiation chemistry of polymers: Effect of radiation on polymer, structure And properties. Application in curing, coating purification, polymer composites etc. radiation induced polymerization.

5. Module 5. Rheology and mechanical properties of polymers (9 hours)

a. Rheology and mechanical properties of polymers: - Introduction to Rheology, Newton's and Hooks laws, rheological response of materials, the ideal fluid, non Newtonian Fluids, time dependent fluids, power law models. Viscous flow, Relationship between stresses and stain, viscoelasticity, Mechanical models – Maxwell and Voight Boltzmann's superposition principles. Kinetic theory of rubber elasticity. The glassy state and the glass transaction, dynamic mechanical testing, relaxation spectrum, frequency dependent visco-elastic behavior stress – strain behavior of elastomers, the mechanical properties of crystalline polymers.

6. Applications of polymers (6 hours)

a. Properties of polymers relevant to surface coatings, printing/painting of plastics, the adhesive and packaging applications.

b. Properties of polymers relevant to the biomedical applications.

c. Properties of polymers relevant to the energy applications.

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3. CHE 4X1 Molecular and Macromolecular Engineering

Prerequisites: CHE 2XX (Molecular Systems in Chemical Engineering)

Objectives: The course is an introduction to polymer science at the graduate and advanced undergraduate level. Topics include polymerization, molecular weight and polydispersity, molecular size and configuration, solution properties, thermodynamics, the glassy and rubbery states, crystallization, viscoelasticity, elastic properties, and multiphase systems.

Textbooks

(1) ???

ABET learning outcomes

1. Ability to apply knowledge of mathematics, science, and engineering to solve material problems related to chemical engineering.

Course learning outcomes

At the end, students will be able to

1. Understand the fundamental and advanced concepts used in macromolecular systems for Chemical Engineers.

2. Know the detailed methods to synthesize and characterize the macromolecular systems

3. Understand the structural, viscoelastic, mechanical and thermal properties of macromolecules and their characterization methods

Course learning outcome assessment methods

1. biweekly homework assignment will assess student's mastery of the basic principles and concepts of macromolecular systems.

2. A mid-term exam and a final exam will assess students' ability to design, characterize, and evaluate macromolecular systems.

3. Four individual papers and One presentation will assess the students' ability of identifying important problems and discussing the challenges and possible solutions.

Purpose of this course in relation to overall curriculum and relationships to other courses offered by primary unit:

This course does not overlap with any other ChE courses. It serves as a core course for Molecular Engineering Concentration. This course is designed to provide more comprehensive knowledge of molecular and macromolecular systems in many emerging fields of Chemical Engineering.

Relationship of this course to similar courses offered by other academic units:

No overlap

Contents (45 hours):

Coming Soon

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4. CHE 4X2 Nanomaterial in Chemical Engineering

Prerequisites: CHE 2XX (Molecular Systems in Chemical Engineering)

Objectives: This course covers the basic principles associated with nanoscience and nanotechnology including the fabrication and synthesis, size dependent properties, characterization, and applications of materials at nanometer length scales with an emphasis on recent technological breakthroughs in the field.

Textbooks

(1) ???

ABET learning outcomes

1. Ability to apply knowledge of mathematics, science, and engineering to solve material problems related to chemical engineering.

Course learning outcomes

At the end, students will be able to

1. Understand the fundamental and essential concepts used in nanomaterials for Chemical Engineering applications.

2. Know the detailed methods to synthesize and characterize the nanomaterials

3. Understand the properties of nanomaterials and characterization methods

4. Design principles of nanodevices and selection criteria for appropriate type of nanomaterial for specific application

5. Offer different approaches to modify structure/microstructure of nanomaterial in order to get desired properties

Course learning outcome assessment methods

1. biweekly homework assignment will assess student's mastery of the basic principles and concepts of molecular systems.

2. A mid-term exam and a final exam will assess students' ability to design, characterize, and evaluate materials.

3. Four individual papers and One presentation will assess the students' ability of identifying important problems and discussing the challenges and possible solutions.

Purpose of this course in relation to overall curriculum and relationships to other courses offered by primary unit:

This course does not overlap with any other ChE courses. It serves as a core course for Nanotechnology Concentration. This course is designed to provide more comprehensive knowledge of nanomaterials in many emerging fields of Chemical Engineering.

Relationship of this course to similar courses offered by other academic units:

No overlap

Contents (45 hours):

1. Physics and Chemistry of Nanomaterials (4 hours)
    a. Nanoscale and atomic-crystal structure
    b. Chemical bonds in Nanomaterials
    c. Techniques of synthesis of nanomaterials
2. Self-assembly and Colloidal Phenomena (6 hours)
    a. Self-assembly of nanomaterials
    b. Nanoparticle Stabilization
3. Solution Chemistry of Nanomaterials (9 Hours)
    a. Quantum Dots (CdS, CdSe, Si, etc)
    b. Metallic Nanoparticles (Ag, Au, Cr, etc)
    c. Insulating Nanoparticles (SiO2, polymer)
    d. Nanowires (Si, Au, Ag)

e. Carbon Nanotubes
f. Nanoribbons
g. Graphene
h. Transition Metal Dichalcogenides
i. Phosphorine, borophene, hBN

4. Electronic Structure of Nanomaterials (7 Hours)
    a. Size-Dependent Chemical & Physical Properties
    b. Quantum Confinement and Electronic Structure
    c. Electrical properties, Conductivity and Resistivity
    d. Electron Transport
    e. Coulomb Blockade in 2D Array of Nanoconductors
    f. Optical Properties of Nanomaterials
    g. Mechanical and Thermal Properties of Nanomaterials
5. Characterization Tools (7 Hours)
    a. AFM
    b. STM
    c. FESEM
    d. Electron Microscopy and Nuclear Magnetic Resonance (NMR)
6. Nano-manufacturing (6 Hours)
    a. Nano thin films and nanocomposites
    b. Photo-Lithography
    c. Nano-Lithography
    d. Large-Scale Synthesis
7. Applications (6 Hours)
    a. Nanostructured Fuel Cells and Solar Cells (1 week)
    b. Nanomaterial Examples – Energy (4 hours)
    c. Nanomaterial Examples – medicine (4 hours)
    d. Nanomaterial Examples – Environmental application (3 hours)
    e. Nanomaterial Examples – other applications (3 hours)

8. Implications, Ethics, & Safety of Nanomaterials (3 Hours)

A. *Program Educational Objectives*

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism and Service	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-Long Learning and Leadership	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

The Program Educational Objectives are prominently listed in the "Objectives and Outcomes" link on the home page of the departmental website: www.che.uic.edu. Additionally, the PEOs are included in all undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the professional capacities represented by the PEOs.

Mapping of ABET Student Learning Outcomes: Old 2014 ABET to New 2020 ABET Outcomes

New Student Learning Outcomes (For ABET 2020)	Old Student Learning Outcomes (ABET 2014)
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(1) Ability to apply knowledge of mathematics, science and engineering (2) Ability to identify, formulate and solve engineering problems
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice (7) Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
(3) an ability to communicate effectively with a range of audiences.	(12) Ability to communicate effectively
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(8) Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context (9) Knowledge of contemporary issues (13) Understanding of professional and ethical responsibilities
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(11) Ability to function on multidisciplinary teams
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(4) Ability to design experiments (5) Ability to conduct experiments (6) Ability to analyze and interpret data

(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	(10) Recognition of the need for and ability to engage in life-long learning

B. Consistency of the Program Educational Objectives with the Mission of the Institution

Component of UIC mission	**Contributing PEO**
To create knowledge that transforms our views of the world and, through sharing and application, transforms the world.	1
To provide a wide range of students with the educational opportunity only a leading research university can offer.	1, 4
To address the challenges and opportunities facing not only Chicago but all Great Cities of the 21st century, as expressed by our Great Cities Commitment.	2, 3
To foster scholarship and practices that reflect and respond to the increasing diversity of the U.S. in a rapidly globalizing world.	3
To train professionals in a wide range of public service disciplines, serving Illinois as the principal educator of health science professionals and as a major healthcare provider to underserved communities.	2

The healthcare component of UIC's mission (arising from the specific history of UIC and its medical school within the Chicago Metropolitan Area) is not directly served by the Chemical Engineering program, and would not be expected among the objectives of typical chemical engineering programs. However, to the extent that chemical engineering impacts on medical technology (e.g., artificial organs, instrumentation, biocompatible materials, pharmaceuticals, etc.) and that graduates could be employed in related sectors, the Chemical Engineering bachelor's degree program could contribute indirectly.

C. Consistency of the Program Educational Objectives with the Mission of the College

Component of College mission	**Contributing PEO**
Educate and train students for careers in innovation and leadership.	1

Expand engineering knowledge through excellence in original research.	1, 4
Develop local partnerships to advance the technology sector in Chicago and Illinois.	2, 3
Provide critical expertise in engineering-related issues to Chicago and Illinois.	1, 2
Serve as a major economic driver to Chicago and Illinois.	1, 2, 3

D. Consistency of the Program Educational Objectives with the Mission of the Department

Component of Department mission	**Contributing PEO**
Serve society through excellence in education, research, and public service.	1,2,3
Provide an education in chemical engineering for our students.	1
Instill in our students the attitudes, values, and vision that will prepare them for lifetimes of continued learning and leadership in their chosen careers.	3, 4
Through scholarship, strive to generate new knowledge and technology for the benefit of the State of Illinois, the nation, and beyond.	2, 3, 4